

Large-vocabulary data-centric dialogues

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Data-centric speech dialogues using large-vocabulary speech have been investigated at BT Laboratories for some time now. The research to date has been focused on applications such as directory access, customer identification, auto-attendant and transaction completion. General lessons have been learned which are applicable to many goal-seeking dialogues where the vocabulary of the task is defined by the data sought. Several trials have been run which have provided evidence of good dialogue design principles. This paper presents a framework for the dialogue design of such services, discussing topics such as mixing spelling with spoken entry and the issues of confirmation and offering of information. In addition, the difficulties of designing for confusions due to homophones and synonyms have also been addressed.

1. Introduction

Telephone-based interactive voice response (IVR) systems have been available for some time in such areas as call control, voice mail, and simple information retrieval tasks. In many cases they have been very successful in reducing the cost of providing telephone services and have consequently enabled a number of new services. Such speech systems are usually based upon TouchTone® (dual tone multi-frequency) telephone keypad entry. This provides the caller with the ability to either perform a menu-style navigation through a structured speech dialogue, or to enter numeric information such as telephone numbers, or account numbers. Recorded speech phrases are usually used as the method to prompt the caller or for presenting information to them. Spoken dialogue techniques for such systems are now well understood.

Developments in speech recognition technology have facilitated the introduction of small vocabulary (tens of words) speech recognition into network-based services. BT CallMinder™, a network-based answering machine, is an example of the effective use of this. It incorporates 'yes/no', 'isolated digit' and 'keyword recognition' into a highly usable service. This replaces the need for callers to remember TouchTone key sequences, which can be daunting for some customers. Dialogues for such systems typically involve navigation of menu structures using command words.

This approach is not appropriate for many applications which involve the entry of information by the user which has a large intrinsic vocabulary. Often these are primarily data entry or search applications requiring the entry of a number of corroborating pieces of information, such as names and addresses. This information is used to search a

database and return further related information or to complete a transaction. This class of application is termed 'data-centric' by the authors and is difficult to design using TouchTone if there are non-numeric input vocabularies. In such cases, large-vocabulary speech recognition must be used instead. Examples of such systems are telephone directory applications, customer identification and flight enquiries. Commercial systems are beginning to be trialled and launched in all of these areas by a number of companies including BT. One such example is the BT Brimstone corporate directory enquiry system [1].

This paper discusses the dialogue design issues surrounding the development of large-vocabulary data-centric services. It focuses on systems which use large-vocabulary, isolated, word-spotting, fluent alpha-numeric or spelling recognisers. However, many of the issues and techniques discussed are equally relevant to fluent speech or 'spoken language systems'. The paper also concentrates on the issues of full automation systems in which operator support may be available but generally only when the caller is getting into difficulty.

2. General interaction model

Many data-centric dialogues, or sub-dialogues, can be modelled using the generalised three phase search and find strategy shown in Fig 1.

The first phase is 'vocabulary entry'. This will involve the dialogue prompting for a number of related vocabulary items to narrow down the search space in order to find the item in question. In a flight enquiry system this might be asking for the airport of departure, airport of arrival, and airline for instance. This phase uses large-vocabulary speech recognition.

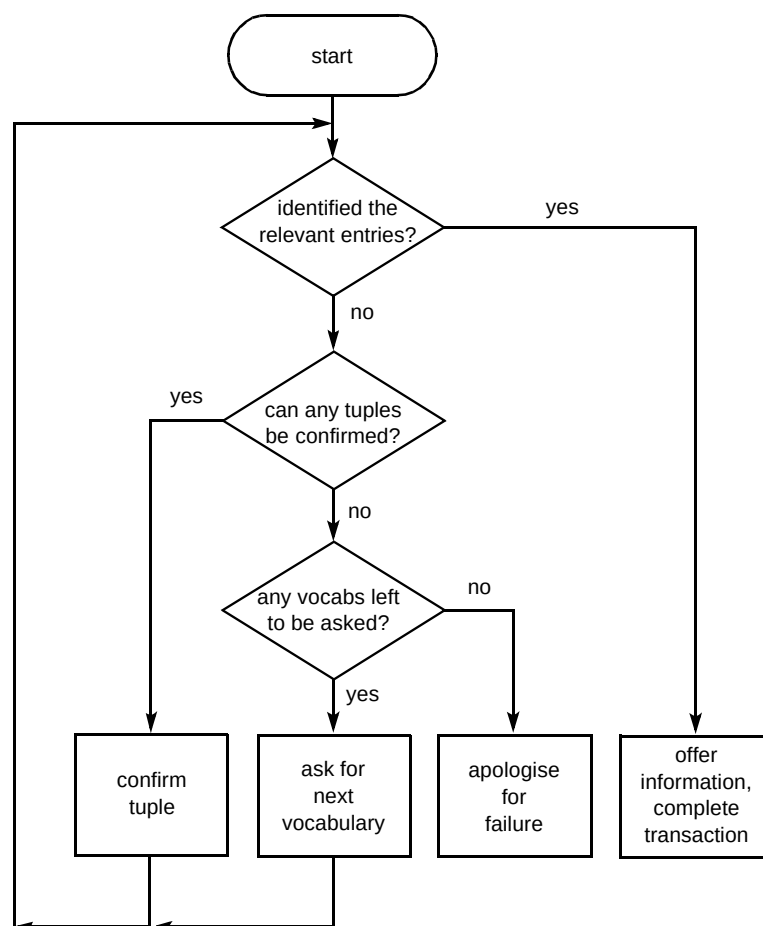


Fig 1 General interaction model.

The second phase is 'confirmation'. This may be the confirmation of information given during vocabulary entry, or the offering of new information for confirmation, which may take the form of disambiguation between similar tuples of information. This phase uses yes/no recognition or small vocabulary speech recognition. Confirmation may be optional if the dialogue is very confident that it has recognised the correct information to date.

The final phase is the 'transaction completion' stage. In this phase, the relevant information has been located, and the caller is able to complete a transaction based on it, or get related information which they do not know.

This phase may comprise the simple offering of an item of information or may be a much more complex sub-dialogue.

The general interaction model of Fig 1 shows how an enquiry may cycle between asking for vocabulary entry and confirming sub-parts of the target entry until sufficient information is gathered to move confidently to the transaction completion stage.

- to confirm items of information as soon as they are entered within the data entry process (mid-query confirmation) which may be tedious for the user and lengthen transaction completion time, or,
- to collect additional information and perform correlation within the information given to perform error repair before entering the explicit confirmation stage (delayed decision making).

Typically, the authors have favoured the latter.

At any stage in the interaction, the dialogue may choose to transfer the caller to an operator. This may be due to poor recognition confidence. It may also be because the enquiry has reached a stage more suited to the use of an operator.

3. Large-vocabulary IVR system architecture

Figure 2 shows a simplified architecture for a data-centric enquiry system. The reader will note that the dialogue and information aspects of the system have been separated.

The recognition strategy component translates strategic dialogue requests into tactical operations. These include interactions with the application database and the

generation and constraining of speech recognition vocabularies. While the recognition strategy tends to be algorithmically complex, many forms of dialogue management can be represented using simple graphical service creation tools. Separating the two allows the iterative design of the user interface to be more independent from the underlying information strategy. This paper is primarily concerned with the dialogue design process but recognition strategies are discussed briefly in section 4.2.2.

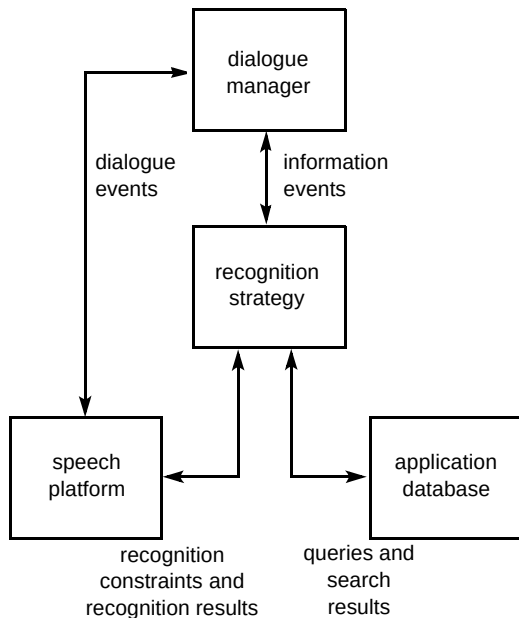


Fig 2 Data-centric architecture.

4. The dialogue design process

The dialogue design of a large-vocabulary IVR application can have a dramatic impact on the success rate of the system given a particular underlying recognition performance [1—3] and consequently its acceptability to callers. This section describes the process of dialogue design which has been found to lead to good data-centric systems.

This process consists of two key phases:

- development of the deep structure of the vocabularies and data
 - vocabulary:
 - finding the natural vocabulary of the application,
 - classifying the vocabularies,
 - defining the intra-vocabulary relationships (vocabulary model);
 - data:
 - defining the inter-vocabulary relationships (data model),

developing the recognition strategy,

defining the criteria which must be met before transaction completion;

- development of the surface structure of the dialogue
 - dialogue structure,
 - dialogue wording.

4.1 Vocabulary analysis

Most data-centric enquiry systems have a number of 'natural' vocabularies. Once found these will form the basis of the dialogue design.

A natural vocabulary is a category of information which a caller will naturally and unambiguously associate with the task. The easiest way to identify these vocabularies is to look at the proper nouns which might be discussed by the caller and the operator in an equivalent manual system, for example:

'What is the **surname** please?'

'Do you know your **account number**?'

'Please give me your **organisational unit code**?'

'Can I have the **telephone number** of my bank?'

'Its **sort code** is 01-25-34'.

Some natural vocabularies are composites; for example, full names can be made up of first names and surnames and phone numbers can be comprised of dialling codes and main numbers. If either the caller or the system needs to discuss these components, for clarity, they should be modelled separately. This is particularly important if the components of the composite vocabulary have separate synonyms or homophones (e.g. Pete/Peter-Smith/Smyth). Issues of vocabulary modelling are discussed further in section 4.1.2.

4.1.1 Classifying vocabularies

Once the natural vocabularies of a system have been identified, the dialogue designer should consider whether each vocabulary is suitable for vocabulary entry and confirmation, disambiguation or used within transaction completion.

During system design, a number of factors must be taken into account for each vocabulary.

- Will the user know what to say to answer the question? (i.e. is it 'Well-known'?)

For example, in the Brimstone system [1] it was felt that users could be expected to know the name of the person but not their alpha-numeric organisational code.

- Will the system be able to predict the user's response? (i.e. is it 'predictable'?)

Many questions can elicit a variety of responses from a caller. In some cases, such as asking for first-names, where a caller may refer to someone who is called Peter in the database as Pete, the ratio is small and the alternatives relatively easy to capture *a priori* (apart from unpredictable nicknames). In others such as department names the relationships may be extremely difficult to capture as the responses may bear little direct similarity to contents of the major data source. Vocabularies which are in some sense canonical, such as credit card numbers, are extremely predictable.

Figure 3 shows how the predictability and the degree to which a vocabulary is 'well known' can guide the dialogue designer to select particular vocabularies for particular tasks. Other considerations for selecting the role of a vocabulary are discussed below.

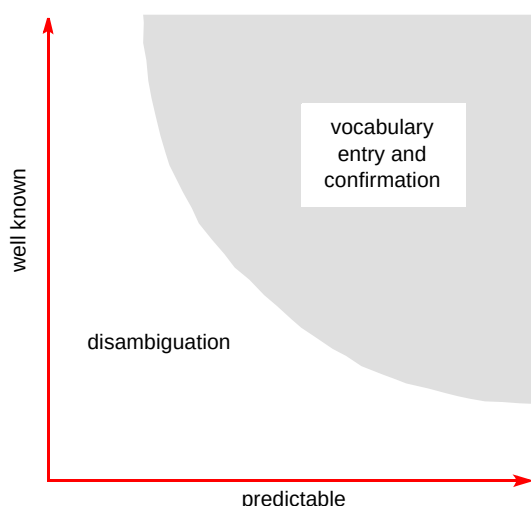


Fig 3 Classification graph for vocabularies.

- Will the user know how to say it, i.e. is it 'natural'?

Typically spoken words are easy for a user, but entry of spelt names can lead to some exceptional cases such as double-barrelled names. Partial spelling "please say the first 3/4/few letters of the ..." is a potential solution but there are still significant dialogue issues associated with this technique.

- Can it be recognised accurately? (i.e. is it 'recognisable'?)

There is a strong inverse relationship between the size of a vocabulary and the recognition performance which can be achieved — this is less true at small sizes where the actual vocabulary is more important. Therefore

large vocabularies, where all vocabulary items are equally likely to be spoken, should be avoided where possible — however the following sections describe techniques for improving effective accuracy on large vocabularies. Typically alphanumeric entry such as spelling and digit string entry can be significantly more accurate for a given vocabulary size than the same spoken vocabulary. The effects of different speakers and channels can also be very significant, e.g. mobile calls and speakers with poor enunciation.

The art of large-vocabulary system design is the balance of these factors, i.e. whether a user knows what to say and how to say it versus our ability to predict the response and recognise it.

Vocabulary ranking

In order to keep the dialogue brief and reliable, the dialogue designer should start by asking for vocabularies which are both well-known and predictable:

- the caller should be able to answer the question,
- the designer should be able to predict the valid responses.

Once all such information is obtained, the designer is forced to rely upon information which is predictable but may not always be known by the caller. Typically, an additional question may be required to find whether the caller knows the particular piece of information, e.g. in a residential directory automation system:

"Please say yes or no. Do you know the road name?"

Additional guidelines apply in the case where spelt and spoken information is combined. These are discussed later in section 4.3.

Confirmation

Any item of information which has been given by the caller can be used in confirmation, for example in a direct yes/no question of the form:

"Please say the surname
Now please say the firstname
Is the name David Attwater?
..."

Attwater
David
Yes

Confirmation is discussed later in more detail in section 4.5.2.

Disambiguation

In some cases, vocabularies within the system are either not well known, poorly predictable or just too difficult to

recognise. However, such vocabularies can still be useful in disambiguating between candidate entries in a database. A number of entries are offered to the user who is required to implicitly or explicitly distinguish between them.

For example, in the Brimstone directory system operational unit names are used to implicitly select between a number of candidate entries.

Where possible, except for those associated with the goal of the discourse, vocabularies which are unlikely to be known by the caller should be avoided within the dialogue.

4.1.2 Designing the vocabulary model

A given vocabulary item, such as a surname, can have a number of different incarnations — for example, it may have a number of valid spellings (e.g. a caller may spell it in a number of ways) and a number of alternative spellings. It is useful to explicitly model these relationships in a vocabulary model.

Figure 4 shows the framework for a vocabulary model as an entity relationship diagram relating together the different representations used for items in data-centric systems. These representations are shown as spoken vocabulary, spelt vocabulary, pronunciations, and concept.

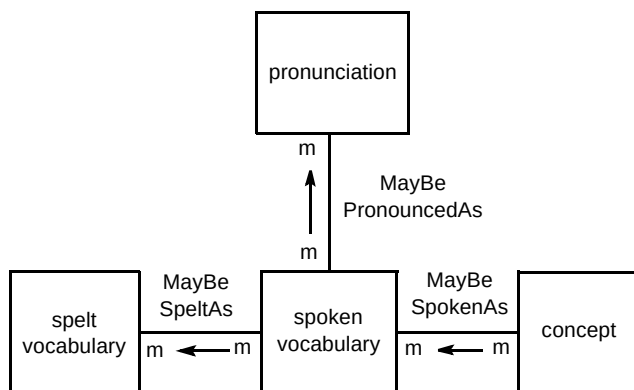


Fig 4 Simplified vocabulary model.

A decision has to be made as to which representations should be used within the various components of the system; for example, should any application database be populated with orthographic or phonetic representations? In order to clarify the representations used, the vocabulary model shown includes an explicit representation of the 'concept'.

The 'concept' is the level at which the vocabulary is related to other vocabularies, in a database for example. These concepts may be used with many different spoken words for a given item. Each spoken item has a spelling, but there may be more than one 'valid' spelling of an item in an application if callers are permitted to spell homophones or

make common spelling errors. For this reason, spelt vocabulary representations are considered to be separate from spoken representations.

Each word will have a pronunciation, but there may be more than one pronunciation for a given word (homonyms), and different words that share pronunciations (homophones).

A vocabulary model should be designed for each vocabulary in the system. This design takes into account the uses that the vocabulary will be put to, which in turn dictates which of the relationships shown in Fig 4 must be defined and populated for the application in question. For example, a vocabulary which will be used for spoken vocabulary entry but not spelt entry will not require the 'MayBeSpeltAs' relationship to be defined. If the vocabulary model is well designed, it should require little change throughout the lifetime of the dialogue.

4.2 Data analysis

Once the vocabulary model is established, the next stage in the dialogue design is to consider the inter-vocabulary relationships in detail.

It has been discovered that in data-centric applications the characteristics of the data in the application have a large impact on the structure of the resulting dialogue. Dialogue design that ignores this fact is likely to fail. The purpose of this stage is threefold. Firstly the relationships between the vocabularies are analysed and a data model produced. Secondly the recognition strategy is identified. This will be used to ensure that the speech recognition is as accurate as possible. Thirdly, a decision must be made about which criteria must be met before transaction completion may occur.

4.2.1 Data model

Establishing the relationships between the vocabularies will yield a data model for the application upon which analysis and implementation will depend. In many systems this data model will contain a single central table or set of rules which relate all of the vocabularies at the conceptual level.

Consider a flight enquiry system, for example. The vocabularies (departure city, departure country, arrival city, arrival country, time of arrival, flight code, airline, etc) may be related by a single table based on the flight schedules. In general, these relationships should be specified between vocabularies at the conceptual level. For example, Virgin Atlantic airlines is a single concept but may be spoken as 'Virgin', 'Virgin Airlines', 'Virgin Atlantic', etc. For

simplicity the relationship of Virgin Airlines to departure cities, for example, should therefore be expressed at this conceptual level, and not at the spoken vocabulary level.

The joint combination of the knowledge base (i.e. database), in which these relationships are defined, and the vocabulary models together form the data model of the system.

4.2.2 Recognition strategy

As has been stated, the purpose of the recognition strategy is to enable the inter-relationships between vocabularies to be exploited to get the highest possible recognition accuracy. This approach has been discussed in some detail elsewhere [2] but will be summarised below.

The data model describes how the different vocabularies are dependent on one another. These dependencies may then be exploited to get increased application recognition performance.

For example, consider the two vocabularies 'departure country' and 'departure city' in the international flight enquiry system. Once it has been established that the departure country is France, there may only be ten or so departure cities in France from which international flights depart. The recognition vocabulary for departure city may thus be constrained to these ten cities, rather than the several hundred around the world, thus dramatically increasing recognition performance.

Rather than confirming each vocabulary as it is entered (mid-query confirmation), it is possible to propagate a number of candidate words forward and, using data obtained later in the dialogue, resolve the ambiguity. This is analogous to the beam search within speech recognition.

Dependent sequences of vocabularies such as these are sensitive to errors within the sequence. There may be two major causes of such errors.

- Recognition errors

Where possible, the designer should attempt to ask for vocabularies with high recognition accuracies early in the sequence — for example, alphanumeric representations, such as spelling and account numbers, often serve the purpose of reliably, but highly, constraining the subsequent vocabularies.

- Domain or prediction errors

It is possible that the database associated with the application may not adequately capture the range of responses which may be encountered from users. For example, in a flight arrival system it may be tempting to constrain the vocabulary of the arrival city given the candidates from the recognition of the departure city. However, this may be sensitive to the case where the caller asks for non-existent routes. The dialogue designer must therefore be sensitive to how likely the user is to respond in a fashion outside of the domain knowledge of the system.

Taking into account these considerations, a recognition strategy can be defined. A framework has been proposed by the authors in which 'tracks' are used to represent the development of these dependent sequences. 'Track algebra' describes how these tracks can be manipulated and combined. At each point a track represents a set of scored database hypotheses and these may, for example, be intersected or compared for consistency (see Fig 5). They may also be considered as a sequence of operations that are dependent upon both the domain knowledge and the uncertain results of recognition to date. Attwater and Whittaker [2] discuss the use of tracks in more detail. Durston et al [4] present a framework for designing such recognition strategies.

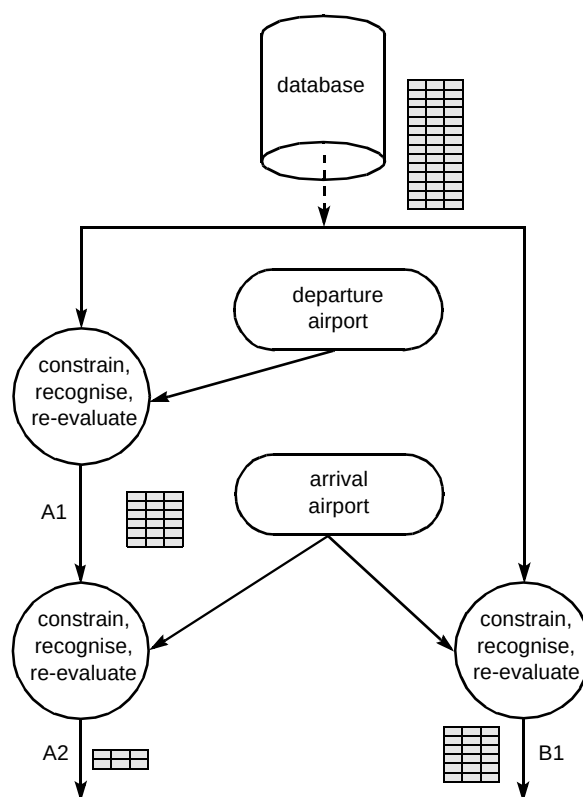


Fig 5 Example portion of a track diagram.

Track algebra provides a powerful approach to detecting and potentially correcting recognition and domain errors. An additional approach is the use of partial vocabulary recognisers (PVRs) in which the vocabulary of the recogniser is constrained *a priori* to the most statistically likely subset of expected responses maximising the effectiveness of traditional rejection strategies. This can be particularly useful in partial automation systems, such as a directory assistance system, which only attempts to recognise frequently requested business listings and forward remaining calls to an operator.

It is often fairly clear which vocabularies will be used for vocabulary entry, and how they are related to one another to ensure maximum recognition performance. This will suggest a preliminary order for questioning the caller.

4.2.3 Transaction completion criteria

A key point in the flow shown earlier in Fig 1 is when the decision is made as to whether enough information has been gathered with sufficient confidence for completion of the dialogue to proceed. The dialogue designer must look at the data model and the task being attempted and decide which criteria must be met before transaction completion may be attempted. Typical requirements may include:

- the identification of a single account in a financial transaction system,
- the identification of a single name corresponding to at most three entries in a directory system such as Brimstone.

Information related to this entry may then be offered to the caller, or modified as part of a transaction.

4.3 Dialogue structure design

The design process for the outline structure of a dialogue will often be iterative. This section discusses the issues involved in designing the structure of the dialogue.

Vocabulary entry order

The recognition strategy will have indicated a desirable question order which will offer the highest recognition performance. It is then important to look at the dialogue and decide what a natural questioning order for the caller will be.

Each application will have unique needs which must be considered. It is particularly important to be careful with question order when the same conceptual piece of information is requested in two different ways (e.g. spoken then spelt).

If an unacceptable question order is suggested by the optimal recognition strategy, then one of two options must be considered. Firstly, alter the strategy, or secondly ask the questions in a different order to processing the results. This can be made possible by recording a response and re-recognising it at a later point in the dialogue. Care must be taken with this approach. Re-prompting due to poor recognition scores becomes difficult to manage in this circumstance.

Confirmation strategy

In addition to explicit vocabulary entry, confirmation may be used to provide an implicit method for gathering additional information. This is represented within the generalised interaction model shown earlier in Fig 1.

It is usually helpful to confirm the caller's request before offering the desired information, or service, related to the request. Such confirmation ensures that recognition errors are detected by the dialogue, and it also helps to orient the caller in preparation for the next phase.

Confirmation enables both the caller and dialogue to proceed with confidence. It may not be necessary to confirm all of the information entered by the user, and it is acceptable to ask for further clarifying information after confirmation has occurred. Consider the flight enquiry system dialogue fragment shown in Fig 6.

Please say the airport at which the flight will arrive	Gatwick
Please say the name of the airline	Virgin
Do you want the details for a Virgin flight arriving	
at Gatwick?	Yes
I have a number of entires matching that information,	
please answer yes or no, do you know the	
departure airport?	Yes
...	

Fig 6 Flight enquiry system dialogue fragment.

In Fig 6, the caller is given confidence that the enquiry is proceeding well, and the system is now sure that the arrival airport is Gatwick and the airline is Virgin. If wrong information is offered at this point, the caller is able to realise that the enquiry is in trouble prior to engaging in further lengthy dialogue to gather extra information.

Confirmation may be explicit using a 'yes/no' question as in Fig 6. It may also be implicit by simply echoing the vocabulary item after it has been spoken.

Experience has shown that it can sometimes be confusing to expect callers to explicitly confirm information which they have not offered during vocabulary entry. For example, the following dialogue (Fig 7) may be confusing to the caller because they are not aware of the alternative airlines available.

Please say the airport at which the flight will arrive *Gatwick*
Is the airline Virgin Atlantic? *Yes*
...

Fig 7 Explicit confirmation of information not offered.

This is a delicate decision which is probably dictated by how well-known to the caller the new vocabulary is.

For vocabularies which are not so well known to the caller, an alternative to confirmation is disambiguation. This may be explicit (asking the user to confirm one of a set of alternatives) or implicit (offering all alternatives with associated information and letting the caller choose which one they require).

Explicit disambiguation might take a form similar to the dialogue fragment in Fig 8.

Please say the city of departure *London*
There are a number of airports in London,
please say Gatwick, Heathrow, London City or
Stansted.
...

Fig 8 Explicit disambiguation dialogue fragment.

Implicit disambiguation is often appropriate during the transaction completion stage of the dialogue.

For example, imagine that the criteria for transaction completion in a corporate directory system demands that less than three people exist with first name and surname which have just been confirmed. In such a case, the alternatives may be offered as part of transaction completion and the caller selects the correct entry for themselves, as in the example in Fig 9.

...
I have three entries matching that information,
The number for Dave Bailey of Networks and Systems is
01473 649329
The number for David Bayley of Facilities and Technical
Management is 01473 645222
The number for David Bailey of Business Operations is
01473 642022
I'll repeat that ...

Fig 9 Sample of a transaction completion.

Mixed modality

There are several modes of data entry which are available to the designer of a large-vocabulary application. Among these are:

- spoken:
 - large vocabulary,
 - alphanumeric;
- spelt;
- TouchTone:
 - numeric string entry,
 - spelling;

- yes/no:
 - confirmation,
 - list selection;
- grunt detection:
 - list selection.

Most of these can be mixed together but a degree of consistency of approach throughout the dialogue is key.

For example, in systems which mix spoken and spelt responses, experiments have shown that callers may be confused with dialogues such as the one shown in Fig 10.

Please spell the surname *F O S T E R*
Now, please say the first name *David*

Fig 10 A mix of spelt and spoken responses.

With this dialogue, callers may have a tendency not to listen closely to the prompts, and use the modality of a previous question to answer the current one. So, in the example in Fig 10, the first name may well be spelt rather than spoken.

Therefore, if spelling is to be mixed with spoken responses, it is advisable to always ask for the spoken word before asking for its spelling even if the spoken response is never used by the system. Thus, the example of Fig 10 would become as shown in Fig 11.

Please say the surname *Foster*
And now, please spell the surname *F O S T E R*
Now, please say the first name *David*

Fig 11 The sample of Fig 10 using spoken response prior to spelling.

Similar issues surround the mixing of TouchTone with other modalities.

A TouchTone entry may be successfully used as a simultaneous alternative to speech input. In these instances, hybrid dialogue prompts such as 'Please say or use the keypad to enter the account number..' should be avoided. The prompt should only ask for a spoken response leaving the TouchTone option as a fast-track option for experienced users who have found out about it from another source. BT's CallMinder is a successful example of this approach permitting the digits '1' and '2' to be used instead of 'yes' and 'no' if desired.

In addition, TouchTone may be preferred where security is required such as PIN entry.

4.4 Dialogue wording

Standard terms for natural vocabularies

The design of messages prompting the caller for vocabulary entry will itself affect the predictability of the response received. The use of clear, closed questions is sufficient to constrain many callers to present concise answers without extraneous speech. A term should be

established for each natural vocabulary in the system, for example 'city of departure'. This term should be used consistently throughout the dialogue. It may also be necessary to establish consistent terms for composite vocabularies, for example the 'whole name' when referring to a firstname/surname pair.

Standard vocabulary entry block

Figure 12 shows a standard block wording for vocabulary entry which has proved reliable in several different applications. The word 'say' is recommended in every block requesting a spoken response and 'spell' for spelt responses. The word 'enter' should be reserved for TouchTone entry.

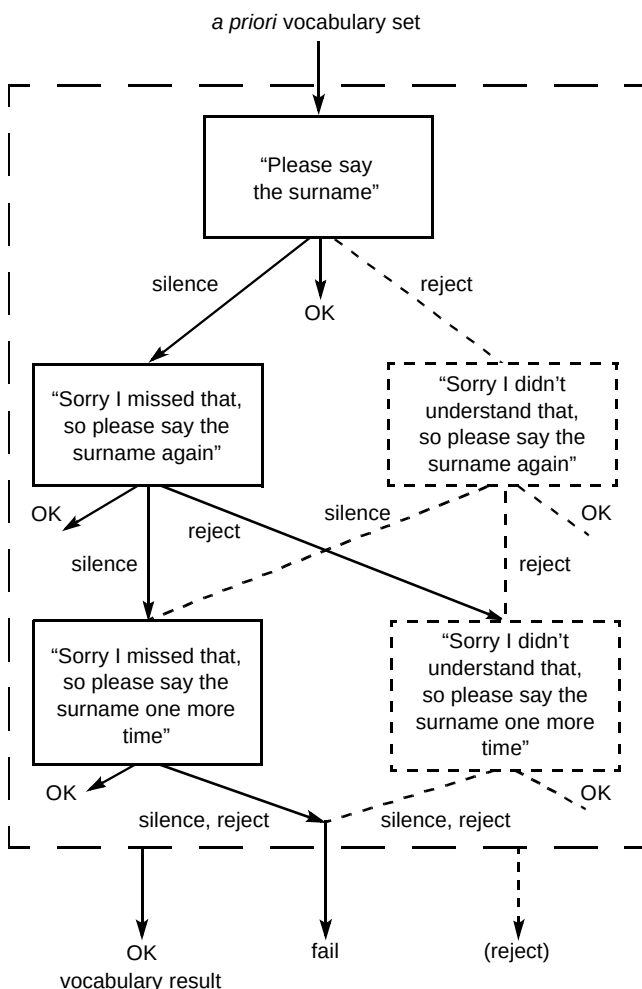


Fig 12 A standard block for vocabulary entry.

The block is primed with the permissible vocabulary set (known as the *a priori* set) and also any probabilities associated with these words if they are deemed necessary to help the recogniser. The resulting output from the block is a vocabulary result (which may be the 'Best *N*' results from the recogniser plus scores) or a status indicating repeated failure due to recogniser rejection or silence.

On occasions it may be desirable to detect rejection at the first level and take other action, for example gaining a corroborative spelling, altering the *a priori* vocabulary set, or forwarding the call to an operator. In this case, the second and third levels need only be defined for silence and not rejection.

How do you know that you know?

Sometimes it is necessary to ask the caller whether they know a particular vocabulary before you ask them for it. It is the current practice of the authors to split this task into two separate questions as in the example in Fig 13.

Please say yes or no, do you know the road name? Yes
Please say the road name? *Lupin Road*

Fig 13 Vocabulary checking.

In this case, if the 'Please say yes or no' is omitted, about half of the callers will respond with the item itself. Exploitation of this effect should be possible.

Confirmation and disambiguation wording

The exact words used for confirmation can be important. The choice of these words can come from two origins. Firstly, the spoken vocabulary, which was originally recognised, could be used directly. This has the advantage of confirming with the user the exact word which was spoken. Alternatively it may be necessary to confirm using a primary spoken word related to a concept which has matched the spoken information. This method has the disadvantage that if there are synonyms for the given vocabulary, the caller may be asked to confirm information using a synonym of the word they originally spoke rather than the word itself.

The first instance is more applicable to implicit confirmation or immediate explicit confirmation of a word, as in Fig 14.

Please say the firstname *David*
Did you say David? *Yes*

Fig 14 Sample of immediate confirmation.

The second instance is more typical when a confirmation occurs later in the dialogue (Fig 15).

Please say the firstname *David*
And now please the surname *Attwater*
Is the name Dave Attwater? *No*
...

Fig 15 Sample of delayed confirmation.

In the example in Fig 15, a synonym is used for the confirmation. In order to cater for this the words 'Did you say' are not used, but instead the caller is questioned as to whether the information they hear is conceptually the same

as what was entered. This is closer to asking what they want rather than what they said.

Distinguishability

It is sometimes desirable to confirm or offer items from a list of potential responses. In these instances, it is not wise to confirm lists of more than three items. In designing systems to date, the authors have taken the approach that the system should not attempt to draw distinctions between things which the user may think are synonymous.

For example, consider the confirmation sub-dialogue in Fig 16.

Please say yes or no, is the name David Bailey? No
Then is the name Dave Bayley ...

Fig 16 Sample confirmation subdialogue.

In this instance, it is inappropriate to expect the caller to draw a distinction between the two surname homophones 'Bayley' and 'Bailey'. It might also be inappropriate to expect the caller to distinguish between the synonyms 'David' and 'Dave'.

In order to avoid this, systems should test whether spoken vocabulary items share common pronunciations or have potential synonym confusions.

If an item is positively confirmed by the caller, all other items considered indistinguishable from that item should also be considered confirmed. This will mean the addition of homophones, and possibly synonyms also, prior to any re-searching the database using the confirmed information.

Coping with confirmation failure

It has been observed that offering a list of alternatives during confirmation is only advantageous when there is a very high chance that the second or third item will contain the correct item. If not, the impact of hearing three incorrect options is even worse than hearing just one item.

Following a failure to get a positive answer in a confirmation or disambiguation, it has been found to be difficult to isolate the error and correct it at this stage. Attempts to do so only confuse the caller further. It is usual practice in this case to place the blame firmly on the system, apologise and give the caller opportunity to try again or transfer them to an operator (Fig 17).

Is the name David Attwater No
Sorry I have found the wrong person.
Please answer yes or no, would you like to make
another enquiry? Yes

Fig 17 Sample of recognition error.

The issue of blame raises an interesting and unexplored issue of personification and role in speech dialogues. In the data-centric dialogue case, there may be some benefit in the dialogue assuming the role of an agent searching a database owned by some other entity. This may help to isolate the issues of blame for recognition error from data quality and coverage problems [5].

4.5 Mixing recorded prompts and text-to-speech (TTS)

Large-vocabulary speech output may be handled in a few different ways. One simple way is to make recordings of each vocabulary item that may need to be spoken out by the dialogue. This is time consuming, and difficult to maintain, but can produce a very high quality interface. This approach is suited to applications whose vocabularies are small to medium sized, and change slowly, if at all. One example of this may be a railway timetable enquiry system where the vocabulary of towns and stations changes very slowly. These towns, and the component words of dates and times may be recorded. Careful concatenation of these recordings in different contexts is able to provide a high-quality interface.

For larger vocabularies, TTS is the only viable alternative. The designer must then choose whether all of the dialogue prompts are to be TTS generated, or simply the variable fields. Many recent TTS systems, including BT's *Laureate*, can be based on a particular individual's voice. This enables TTS generated variable fields to be spliced into recorded sentences from the same speaker. Care must then be taken with the prosody and intonation of the resulting sentences.

Recent experiments have indicated that mixing TTS recorded words with pre-recorded carrier phrases is of higher quality than a completely TTS generated dialogue [6]. However, trials have also shown that this mixing is still a key area of dissatisfaction for callers. Some interesting approaches have been proposed for this problem [7] but more research is required.

5. Conclusions

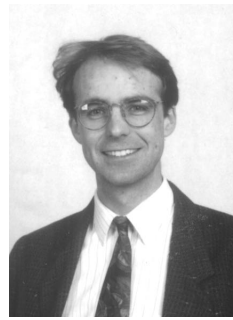
It is now possible to build data-centric services with large-vocabulary speech recognition. In order to make these services accurate and usable, the dialogue design process must consider carefully the information processing requirements of the application.

In order to do this, the natural vocabularies of the domain must be found, analysed and then conceptually related to one another. A dialogue structure must then be found which enables a natural interaction to achieve the highest possible application performance. Once this structure is established, the exact wording of the dialogue must be carefully considered. Issues surrounding spelling

and the mixing of text to speech with recorded speech are still not resolved and require further investigation.

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